Raunit Kohli

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WORK EXPERIENCE

Rocket Software **Boston, MA**

Machine Learning Software Engineer III 09/2024 – Present

- Owns core features of Rocket EVA™ AI platform: sanitized RESTful API, end-to-end authorization, streaming telemetry, async MCP tool execution engine, multi-database Docker deployment, and live SMF record analysis tooling.
- Creates and delivers AI foundations workshops for company apprenticeship program at global offices (65+ graduates), achieving 100% content adoption; leading coordination of enterprise-wide AI enablement program for 1800+ engineers.
- Built SAS-to-Python code translation tool on specialized team using DSPy ReAct agent, collaborating on Agentic Context Engineering (ACE) optimization framework; created multi-agent workflows for architecture design, test generation, and automated code execution refinement with MLflow evaluation.
- Designed and implemented novel Large Language Model Citation Generator for SmartChat™ using multi-Gaussian clustering algorithms and dynamic content similarity techniques, yielding 80% more relevant document references.
- Developed few-shot LLM prompting framework to translate natural language queries into executable SQL, integrating multi-step sub-agent workflows and regex-based syntax validation, enhancing table-based document retrieval and QA.

Robotics Using Bayesian Inference Lab – UC San Diego **San Diego, CA**

Machine Learning Research Consultant (Published) 09/2024 – Present

- Architected composable spatial and neural signal analysis pipeline and presented IEEE ICDL 2025 publication on explore-exploit dynamics in rat-robot social interactions, characterizing homebase-derived proxemic features for social behavior.
- Developing bio-influenced deep RL regulation framework in MuJoCo simulation using multiple policy optimization algorithms to model exploration-regulation dynamics in autonomous robotic behavior.
- Founding engineer hired for multi-million dollar AI-powered classroom project; built PTPv2-synchronized multi-modal embedded system for secure data streaming and recording with custom Arduino firmware for OpenBCI biophysical sensors and high-precision Basler camera integration.

Staff Machine Learning Research Engineer (Published) 01/2021 – 09/2024

- Spearheaded implementation of Reinforcement Learning algorithms in PyTorch for robotic PiRat over ROS2 with sim-to-real transfer of autonomous behavioral routines to physical robot velocity commands, utilizing extensive neural and positional data from 2500+ rat experiments to mimic and predict multi-agent interactions.
- Developed live multi-agent pose estimation system using SLEAP neural networks with multi-threaded architecture.
- Enhanced and scaled production-deployed Computer Vision-based live Pose-Estimation Software in C++ to validate statistical accuracy of Kinect Tracker, introducing new features and parameter logging, yielding 3x processing speed.
- Developed signal processing applications for live and post analysis, leveraging advanced filtering techniques to enhance identification of stress-inducing behaviors from neural and physiological signals (HRV, PPG) of project subjects.

HIGHLIGHTED PROJECTS

VLM-Guided Hierarchical RL in Habitat-Lab (WPI, 2025) – Hierarchical PPO/SAC for Meta Research's Fetch robot with VLM high-level planner for autonomous navigation and manipulation from language instructions; details.

Smart Scrum Master – Rocket.Build 2025 "Most Innovative" Winner – LLM-powered Agile assistant and MCP Server optimizing Jira workflows via reasoning agent, winning 1 of 6 awards from 650+ global company projects; details.

Aspera Workflows Automated Test Suite (IBM Engineering Intern, 2023) – Built complete 780+ test automated regression framework in Ruby for IBM Aspera on Cloud API, identifying 15+ critical production bugs across all endpoints.

VEX Robotics Autonomous Systems Coach (Grafton High School, MA) – Mentoring competitive robotics team in autonomous path planning: PID control, spline trajectory generation, odometry-based localization, and sensor fusion.

TECHNICAL SKILLS

Languages: Python, C++, Ruby, Java, SQL, Bash; **Libraries:** PyTorch, DSPy, LangChain, HuggingFace, CUDA, OpenCV
Domains: Policy-Optimization Algorithms, Multi-Agent Systems, Agentic AI, Natural Language Processing, Computer Vision, Markov Decision Processes, Signal Processing, Pose Estimation, Data Analysis (Pandas/NumPy/Scikit-learn)
Tools: AWS, Docker, Kubernetes, Postgres, Kafka, ROS2, MuJoCo, Git/GitHub, Atlassian (Jira, Confluence, Bitbucket)

EDUCATION

University of California – San Diego **09/2020 – 12/2023**

Bachelor of Science, Machine Learning and Neural Computation Major GPA: 3.91/4.0

Minor: Computer Science; **Program of Concentration:** Philosophy

Awards: President & Recruitment Advisor – Phi Sigma Pi National Honor Fraternity; Provost Honors (Warren College)